

DevOps Capstone Project 2

Healthcare Appointment Management Platform

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18-MAY-2026

1. Technologies Used:

- Trello
- GitHub
- GitHub Actions
- Python Flask
- Pytest

2. Project Overview

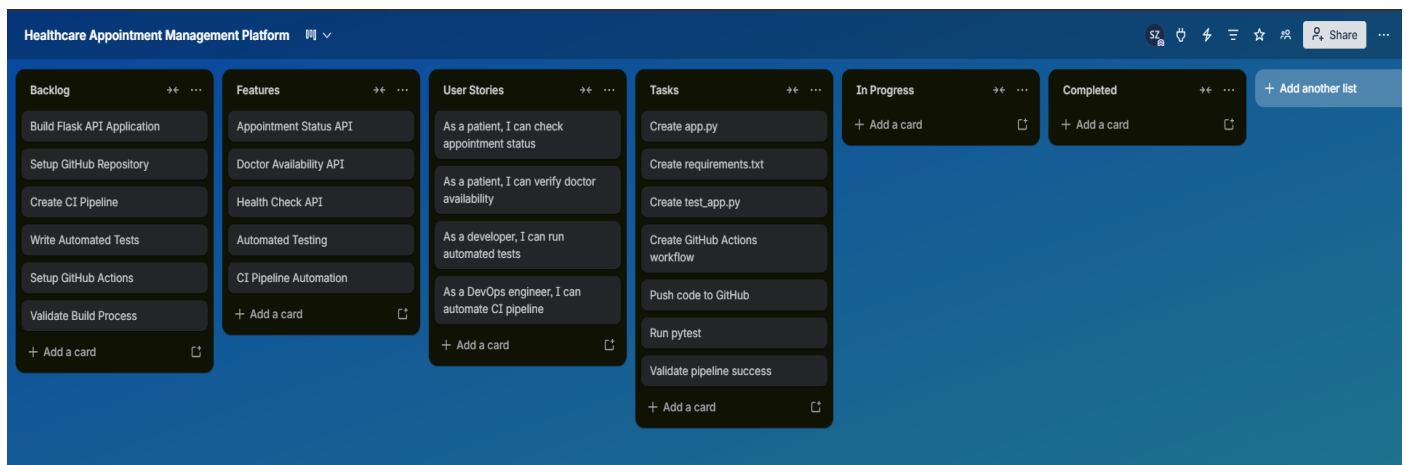
This project demonstrates a DevOps-based CI workflow for a Healthcare Appointment Management Platform using Agile planning, source control management, automated testing, and CI pipeline automation. The application provides APIs for:

- Health Check
- Patient Appointment Status
- Doctor Availability Status

The project uses GitHub Actions to automatically run tests whenever code is pushed to the repository.

3. Agile Planning using Trello

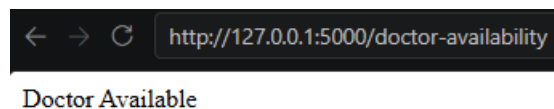
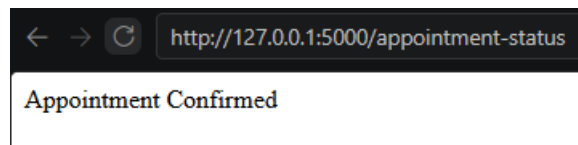
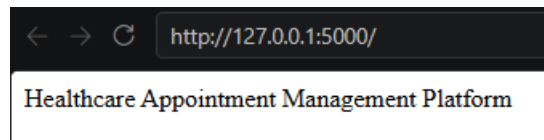
A Trello board was created to manage Agile planning activities including Features, User Stories, asks, and workflow tracking.



4. Flask Application Development

A Flask-based Python application was developed to expose APIs for:

- Home Page
- Health Check
- Appointment Status
- Doctor Availability



5. Application Code

The application was developed using Python Flask framework and structured using modular files including:

- app.py
- requirements.txt
- test_app.py

6. Testing using Pytest

Automated testing was implemented using Pytest to validate all API endpoints and ensure successful responses.

```
PS C:\Users\zerub\OneDrive\Desktop\healthcare-appointment-management-platform> python -m pytest
===== test session starts =====
platform win32 -- Python 3.11.9, pytest-9.0.3, pluggy-1.6.0
rootdir: C:\Users\zerub\OneDrive\Desktop\healthcare-appointment-management-platform
collected 4 items

test_app.py .... [100%]

===== 4 passed in 0.39s =====
```

7. Source Control using GitHub

The project source code was pushed to GitHub repository for version control and collaboration.

```
PS C:\Users\zerub\OneDrive\Desktop\healthcare-appointment-management-platform> git init
Initialized empty Git repository in C:/Users/zerub/OneDrive/Desktop/healthcare-appointment-management-platform/.git/
PS C:\Users\zerub\OneDrive\Desktop\healthcare-appointment-management-platform> git add .
PS C:\Users\zerub\OneDrive\Desktop\healthcare-appointment-management-platform> git commit -m "Initial healthcare appointment management application"
[master (root-commit) 5bdc22e] Initial healthcare appointment management application
 6 files changed, 45 insertions(+)
 create mode 100644 .github/workflows/python-app.yml
 create mode 100644 __pycache__/app.cpython-311.pyc
 create mode 100644 __pycache__/test_app.cpython-311-pytest-9.0.3.pyc
 create mode 100644 app.py
 create mode 100644 requirements.txt
 create mode 100644 test_app.py
PS C:\Users\zerub\OneDrive\Desktop\healthcare-appointment-management-platform> git remote add origin https://github.com/ShinyZerubbaRajiveS/healthcare-appointment-management-platform.git
PS C:\Users\zerub\OneDrive\Desktop\healthcare-appointment-management-platform> git branch -M main
PS C:\Users\zerub\OneDrive\Desktop\healthcare-appointment-management-platform> git push -u origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 12 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (11/11), 2.60 KiB | 106.00 KiB/s, done.
Total 11 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ShinyZerubbaRajiveS/healthcare-appointment-management-platform.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\zerub\OneDrive\Desktop\healthcare-appointment-management-platform>
```

The screenshot shows the GitHub interface for the repository 'healthcare-appointment-management-platform'. The repository is public and owned by 'ShinyZerubbaRajiveS'. It has 0 stars, 0 forks, and 0 watchers. The main branch is selected. The file list shows the following files: .github/workflows, __pycache__, app.py, requirements.txt, and test_app.py, all committed 1 minute ago. The 'About' section describes it as a 'DevOps Capstone Project for Healthcare Appointment Management'. There are no releases published.

8. CI Pipeline using GitHub Actions

GitHub Actions workflow was created to automate:

- Dependency installation
- Build validation
- Automated testing

The workflow automatically triggers whenever code is pushed to the main branch.

9. GitHub Actions Workflow File

Workflow File:

.github/workflows/python-app.yml

10. CI Pipeline Execution

The CI pipeline executed successfully in GitHub Actions and all automated tests passed successfully.

```
PS C:\Users\zerub\OneDrive\Desktop\healthcare-appointment-management-platform> git add .
PS C:\Users\zerub\OneDrive\Desktop\healthcare-appointment-management-platform> git commit -m "Added GitHub Actions CI pipeline"
[main 99508c7] Added GitHub Actions CI pipeline
1 file changed, 27 insertions(+)
PS C:\Users\zerub\OneDrive\Desktop\healthcare-appointment-management-platform> git push
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 642 bytes | 214.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/ShinyZerubbaRajiveS/healthcare-appointment-management-platform.git
5bdc22e..99508c7 main -> main
```

The screenshot shows the GitHub Actions interface for the workflow 'python-app.yml' in the repository 'ShinyZerubbaRajiveS / healthcare-appointment-management-platform'. The workflow is titled 'Added GitHub Actions CI pipeline #2' and is in a 'Success' state. It was triggered by a push to the 'main' branch. The summary shows that the workflow ran successfully, with a total duration of 16s. The workflow file is displayed as 'python-app.yml on: push', and the build-and-test job is shown as completed in 11s.

11. Final Trello Board

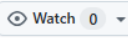
The screenshot shows a Trello board titled 'Healthcare Appointment Management Platform'. The board is organized into six columns: Backlog, Features, User Stories, Tasks, In Progress, and Completed. The 'Features' column contains cards for 'Appointment Status API', 'Doctor Availability API', 'Health Check API', 'Automated Testing', and 'CI Pipeline Automation'. The 'User Stories' column contains cards for 'As a patient, I can check appointment status', 'As a patient, I can verify doctor availability', 'As a developer, I can run automated tests', and 'As a DevOps engineer, I can automate CI pipeline'. The 'Completed' column contains a long list of tasks, including 'Build Flask API Application', 'Setup GitHub Repository', 'Create CI Pipeline', 'Write Automated Tests', 'Setup GitHub Actions', 'Validate Build Process', 'Create app.py', 'Create requirements.txt', 'Create test_app.py', 'Create GitHub Actions workflow', 'Push code to GitHub', 'Run pytest', and 'Validate pipeline success'.

12. Final GitHub Repo



healthcare-appointment-management-platform

Public



main

1 Branch 0 Tags

Go to file

T

Add file

Code



ShinyZerubbaRajiveS Added GitHub Actions CI pipeline

99508c7 · 2 minutes ago 2 Commits

.github/workflows	Added GitHub Actions CI pipeline	2 minutes ago
__pycache__	Initial healthcare appointment management application	4 minutes ago
app.py	Initial healthcare appointment management application	4 minutes ago
requirements.txt	Initial healthcare appointment management application	4 minutes ago
test_app.py	Initial healthcare appointment management application	4 minutes ago